

EXP 4: Build an Artificial Neural Network by implementing the Backpropagation algorithm and test the same using appropriate data sets. (K.Kameshwaran 192312046)

CODE:

```
# Import libraries from sklearn.datasets import load_iris from
sklearn.model_selection import train_test_split from
sklearn.preprocessing import StandardScaler from
sklearn.neural_network import MLPClassifier from
sklearn.metrics import accuracy_score, classification_report

# Load dataset
iris = load_iris()
X = iris.data y =
iris.target

# Split dataset
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# Normalize data scaler
scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)

# Create ANN model model
model = MLPClassifier(
    hidden_layer_sizes=(10,),
    activation='relu',
    solver='adam',
    max_iter=1000,
    random_state=42
)

# Train model (Backpropagation)
model.fit(X_train, y_train)
```

```
# Predict y_pred =
model.predict(X_test)

# Accuracy print("Accuracy:",
accuracy_score(y_test, y_pred))

# Classification Report print("\nClassification
Report:") print(classification_report(y_test,
y_pred))
```

OUTPUT:

Accuracy: 1.0

Classification Report:

	precision	recall	f1-score	support
0	1.00	1.00	1.00	10
1	1.00	1.00	1.00	9
2	1.00	1.00	1.00	11
accuracy			1.00	
30				
macro avg	1.00	1.00	1.00	30
weighted avg	1.00	1.00	1.00	30

```
... Accuracy: 1.0

Classification Report:
              precision    recall  f1-score   support

     0       1.00      1.00      1.00        10
     1       1.00      1.00      1.00         9
     2       1.00      1.00      1.00        11

 accuracy          1.00          30
 macro avg          1.00          30
 weighted avg          1.00          30
```